

BTEC Level 3 Applied Science · Unit 1 Physics · Answer sheet

Progressive Waves — Answers

Mark scheme for the companion worksheet. Accept equivalent correct wording.

A

1. A: max displacement from equilibrium. λ : shortest distance between points in phase. f: oscillations per second.
2. $v = f\lambda$; m s^{-1} , Hz, m.
3. $T = 1/f$.

B

4. Transfers energy (and information), not a net transfer of matter.
5. Frequency.
6. Vertical distance from the equilibrium line to a crest (not crest-to-trough).

C

7. 3.2 m s^{-1} .
8. $f = 2.00 \times 10^8 \text{ Hz}$; $\lambda = 1.50 \text{ m}$.
9. $v = 300 \text{ m s}^{-1}$; $f = 150 \text{ Hz}$.

D

10. See retrieval definitions.
11. $\lambda = 340/510 = 0.667 \text{ m}$ (3 s.f.).
12. Particles oscillate about equilibrium; energy is passed along; no net movement of air.

E

13. $T = 20 \text{ ms} / 5 = 4.0 \text{ ms}$; $f = 250 \text{ Hz}$; $\lambda = 0.040 \text{ m}$; $v = 10 \text{ m s}^{-1}$.